Mobile Robot Challenge: Visual Object Following

Live Demo on May 1st 2026, Van Steenis F1.04

The main goal of this Mobile Robot Challenge is to program the mobile robot such that it is capable to detect an object and follow it keeping a constant distance to the object. The object is connected to a thin rope of length 1 to 1.5m, which is used to tow the object around, following a loosely predetermined trajectory.

1. The above challenge is executed using 1 object of your own choice, dragged along a trajectory of your own choice starting and ending at the same location.
2. The above challenge is executed using 1 unknown object dragged along a given trajectory around two other much bigger objects that will be placed at certain fixed locations.

Important notes:

- **Home Position:** the home position will be clearly marked.
- The robot should face forward at the Home Position.
- The only allowed sensor data used by the robot are its RGB camera data.
- The given object is dragged along, so it will not have a static orientation relative to the robot.
- The given object will always touch the floor.
- It is not allowed to attach the object to the robot.
- You are not allowed to take your own floor, or your own light with you.

In the following two sections Challenge I and Challenge II are described. In the last section the deliverables and grading criteria are described.

Challenge I: Autonomously follow a known object along a trajectory of your choice.

Challenge I is depicted in Figure 1.

1. Place your robot on the designated Home Position such that the camera is facing forward.
2. Your robot has at most 1 minute to capture and process the environment.
3. Place your object within the field of view of the robot.
4. Start your program enabling your robot to autonomously follow the object while you drag it along the trajectory you planned.
5. Let the robot autonomously follow your object following your trajectory starting and ending at the Home Position

NB For this challenge the robot is only allowed to use its RGB Camera sensor! You are not allowed to use its ultrasound sensor, or any other internal or external sensors.

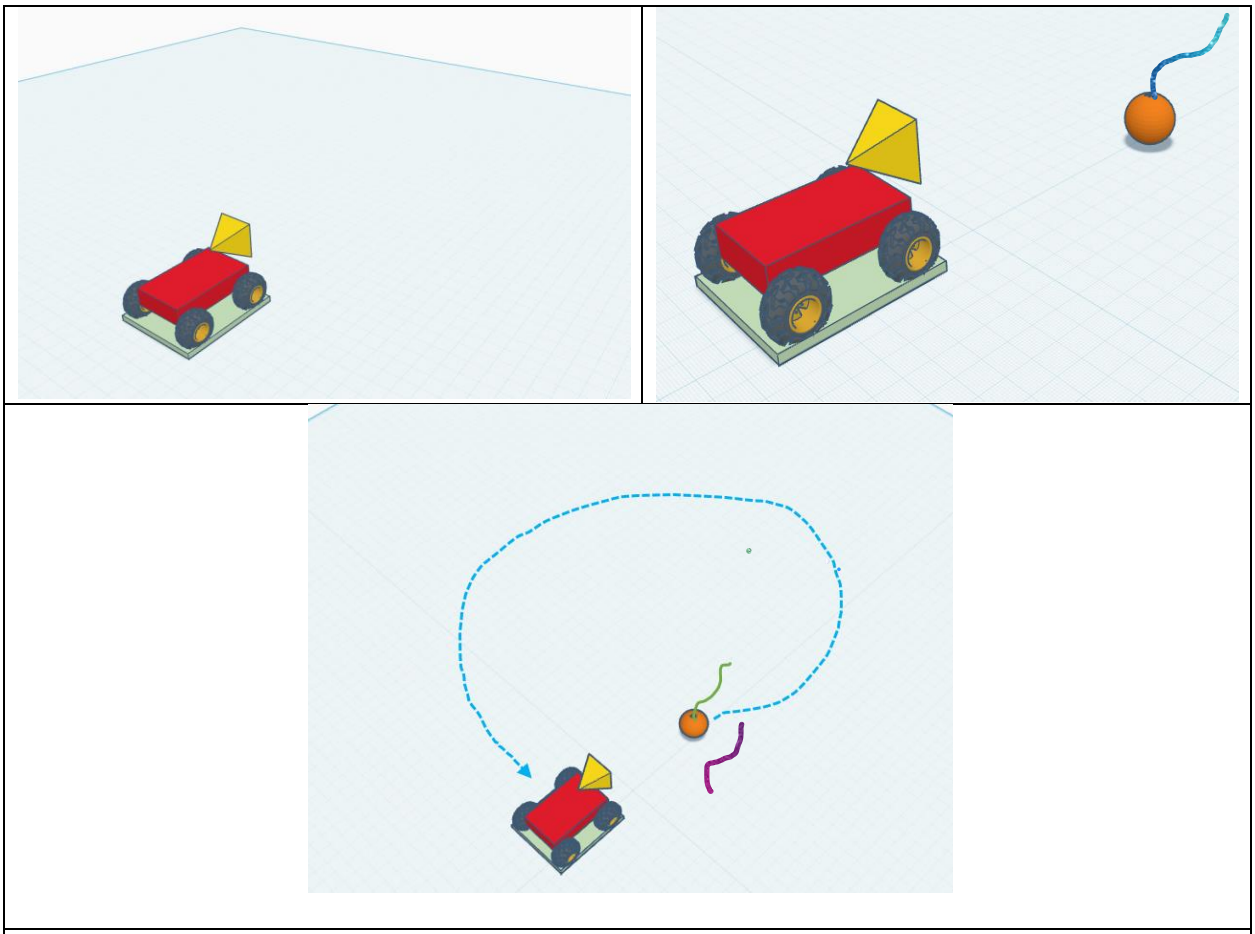

Figure 1: The *left upper image* shows Step 1 of Challenge I. The right upper image shows Steps 2 – 3. The lower image shows Steps 4 – 5 of Challenge I. Note, object and trajectory are just examples.

Challenge II: Autonomously follow a given object along a given trajectory.

Challenge II is depicted in Figure 2.

1. Place your robot on the designated Home Position such that the camera is facing forward.
2. Your robot has at most 1 minute to capture and process the environment.
3. Place the given object within the field of view of the robot.
4. Start your program enabling your robot autonomously follow the object while you drag it along the given trajectory.
5. Let the robot autonomously follow the given object following the given trajectory starting and ending at the Home Position

NB For this challenge the robot is only allowed to use its RGB Camera sensor! You are not allowed to use its ultrasound sensor, or any other internal or external sensors.

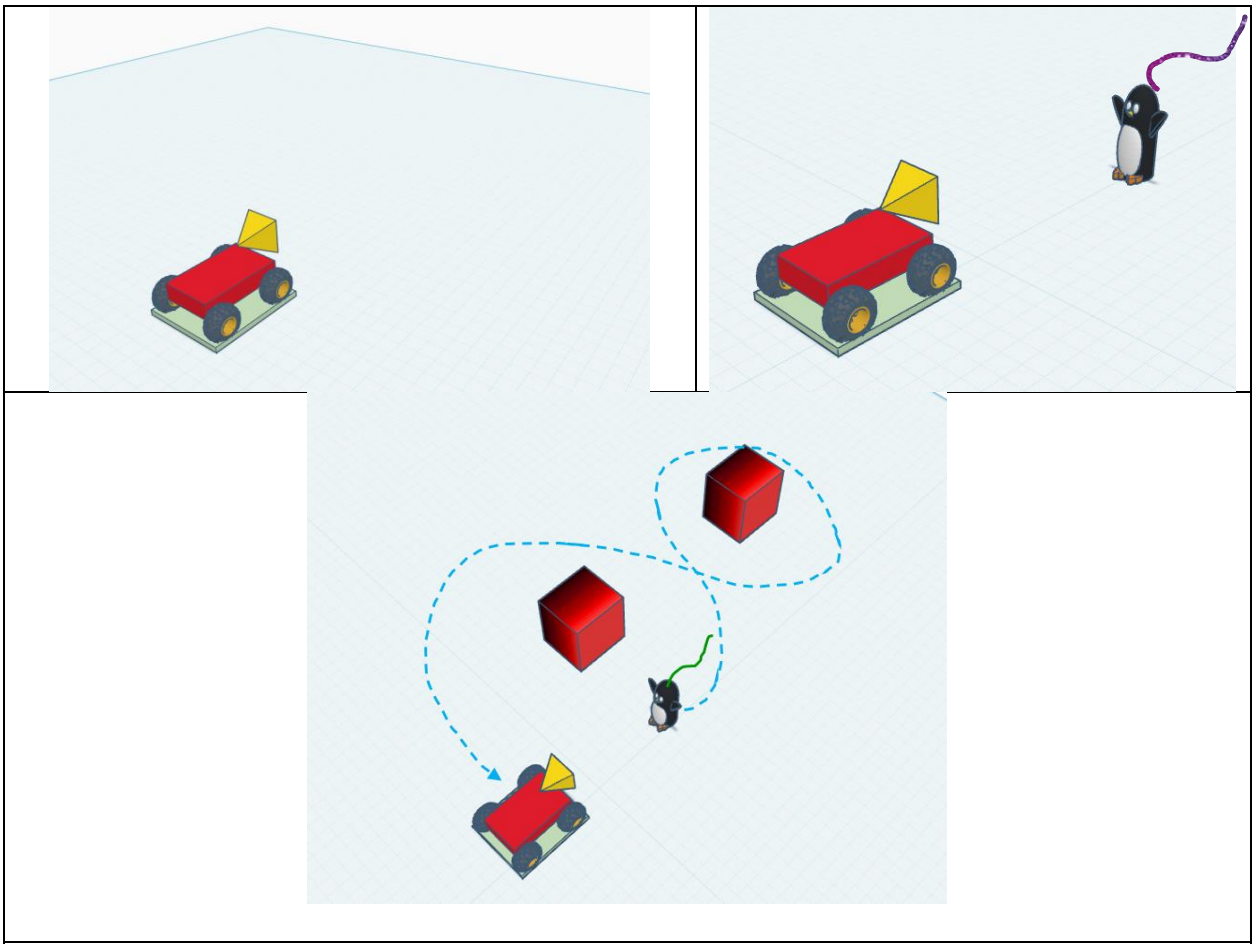

Figure 2: The *left upper image* shows Step 1 of Challenge II. The right upper image shows Steps 2 – 3. The lower image shows Steps 4 – 5 of Challenge II. Note object and trajectory are just examples.

Deliverables:

1. On May 1st 2026 every team should show during a live demo that the robot is able to pass the two challenges.
2. The mobile robot challenge project files are due May 11th 2026. Submit on Brightspace a zip-file containing:
    a. Your source-code.
    b. A short movie, or a link to it, of your robot executing the first challenge.
    c. A pdf with a technical report (max 4 pages, font-size 12) containing a title, the name of the team, a list of all the team members, a description and explanation of the methods used to solve the problems of the challenge, and references to any libraries and papers used. Use a scientific writing style. Finally, in Appendix I give the instructions of any dependencies of your code (i.e., extra libraries that need to be installed to get your code running) and how to execute your code to pass the two challenges.

Grading:

The **final grade** will be based on the originality, speed, smoothness and robustness of your methods and solutions (note, it is perfectly fine, if you focus on a sub selection of this set of characteristics; please clearly indicate that in your report), the performance during the challenge, the quality of your code, the performance captured in your own movie, and the (scientific) quality of the technical report.